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REVIEW ARTICLE

A review of advanced cooling methodologies for solar photovoltaic and thermoelectric hybrid energy systems

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Abstract

Solar energy has several benefits compared to other renewable energy sources, including ease of accessibility and improved predictability. Heating, desalination, and electricity production are a few applications. The cooling of photovoltaic thermoelectric (PV-TE) hybrid solar energy systems is one method to improve the productive life of such systems with effective solar energy utilization. This review critically analyzes the current cooling technologies' various cooling methods and scope. The cooling methods are primarily air-cooling, liquid cooling, nanofluids, and phase change materials. The distinct factors governing the systems' performance were efficiency, coefficient of performance, exergy, and exergy efficiency. Combining active and passive cooling technologies results in a higher PV cell temperature reduction with enhanced PV efficiency. Forced cooling is more productive by about 30% than natural cooling but is not cost-effective. Experiments conducted with nanofluid increased the exergy generation to a great extent by up to 90%. In addition, related issues, such as overheating PV-TE generators, are discussed and recommendations for future research studies are provided. This review benefits all stakeholders of the solar community by designing effective heat exchangers for solar energy systems.

Keywords Solar energy · Solar PV · Thermoelectric generator · Phase change materials · Natural and forced cooling

Abbreviations

COP	Coefficient of performance
CPV	Concentrated photovoltaic
PCM	Phase change material
PV	Photovoltaic
PVT	Photovoltaic thermal

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PV-TE Photovoltaic thermoelectric
TE Thermoelectric
TEG Thermoelectric generator

1 Introduction

Cooling technologies benefit several engineering applications to improve the energy systems' overall performance and productive life. Air, liquid, nanofluids, and phasechange materials (PCM) are the mediums. Recently, cooling technology advances have sustained the global solar energy and electric vehicle battery market. Thus, the cooling of PV-TE hybrid solar energy systems is one of the effective methods to improve the productive life of such systems with effective solar energy utilization. The distinct factors governing the systems' performance were efficiency, coefficient of performance, exergy and exergy efficiency. This present review critically analyzes the various cooling methods and the scope of the current and advanced cooling technologies.

1.1 Sustainable development

Sustainable development goals focus on restoring the earth's health and providing a healthier environment for future generations. Effective utilization of renewable energy sources and minimizing fossil fuel dependency are beneficial. Greenhouse gases are considered to harm the environment because they increase global warming alarmingly. They include major carbon dioxide constituents, nitrogen oxides, hydrochlorofluorocarbons, and methane. The production of such gases can be avoided by encouraging users to shift towards renewable energy sources such as solar, wind, and tidal energy. Solar PV systems are successful due to minimal maintenance. However, lower electrical efficiency occurs from overheating panels by high ambient temperature and dust accumulation. Increasing every one degree Celsius of panel temperature decreases the efficiency by 0.25–0.5% [1]. Electricity production of PV is improved because of the cooling of the PV system [2]. External power provides cooling for active cooling, and passive methods do not require external power [3, 4].

1.2 Motivation for PV cooling

Solar energy conversion to electricity occurs in direct and indirect methods. The direct methods are photovoltaic (PV) and PV-TE. The indirect methods include parabolic troughs, linear Fresnel reflectors, solar power towers, panels, and dish systems [5, 6]. Researchers faced various challenges installing PV systems, such as high cost but lower efficiency. This persuaded them to work extensively on cooling the PV system [7, 8]. The cooling cost is another concern while calculating the efficiency as adequate cooling is to be given by the system to compensate for the overall loss in electricity consumed by the cooling. A PV-TE cooling system can provide cooling in hot and arid regions of the Earth [9, 10].

1.3 Previous work and inventions

The types of cooling are active cooling and passive cooling of PV modules. Active cooling includes forced cooling using pumps/fans. Passive techniques include natural air convection by increasing the exposed surface area with the help of design alterations such as fins, water immersion, conductive cooling using heat sinks, and other high thermally conductive materials. Passive techniques are more accessible and more cost-effective to produce and use than active techniques. These techniques are essential to incorporate environmental concerns that affect the performance of the PV system. Such concerns are sunlight, surface temperature, wind speed, the moisture content in the environment, fine particles such as dust, and the positioning of panels, which enormously affect the system's efficiency [11].

1.4 Current and further scope

The hybrid PV-TE system is a novel cooling system widely researched. A thermoelectric (TE) device can directly convert excess thermal energy into electrical energy through the temperature difference between the two faces of a PV panel [12]. PV-thermal systems are available in highly integrated configurations, such as PV with PCM. PV-thermal systems are based on nanofluid and PV-TE. Amongst all available systems, the PV integration with TE has grown promisingly due to its unvarying results across various recent applications [13]. Among all the available systems that used natural air-cooling as a means of cooling, only three performance methods could be compared since the efficiency values are provided. Figure 1 shows the types of cooling of PV.

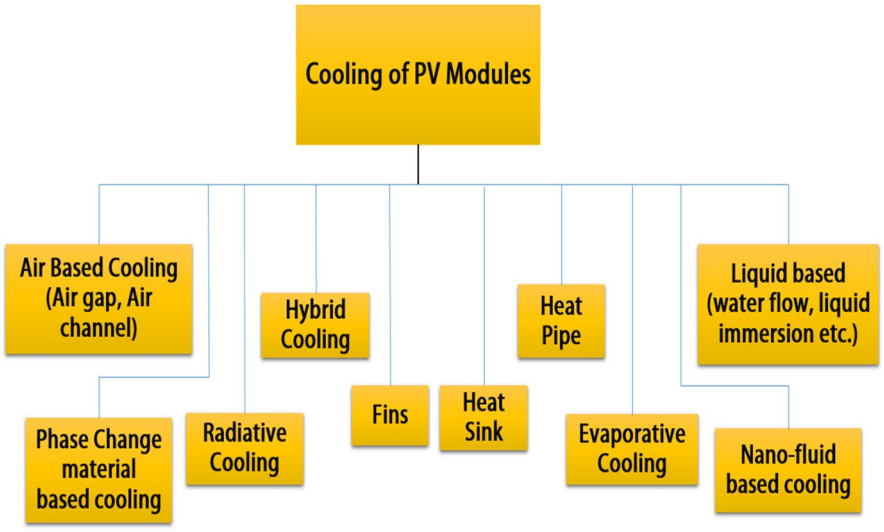


Fig. 1 Types of cooling of PV

Table 1 Comparison of earlier reviews on PV-TE

References	Content of the review article
[14]	This review analyzes various hybrid PV-TE technologies of excess heat management of PV modules from articles published before 2019
[15]	This review overviews the thermal management of photovoltaic thermal (PVT) collectors, heat pumps, and buildings applications wit
[16]	This review proposes using unsegregated water resource management in concentrated solar energy infrastructure plans with articles published before 2020
[17]	This reviews design modification on solar still's performance and for domestic and industrial applications with publications before 2021
[18]	This review studies the concept of PV-TE devices using models and numerical investigations to achieve higher efficiency with recent works published before 2021
[19]	This review article portrays various integration options of the concentrated PV- TE system and different hybrid systems with publications published before 2021
[20]	This article studies a detailed 3D finite element approach to the efficiency enhancement of hybrid systems using passive techniques, with the literature before 2022
The present review	The present review comprehensively deals with the recent development and advances in cooling technologies of PV-TE with the recently published articles, especially 56% literature, from 2020 to 2022

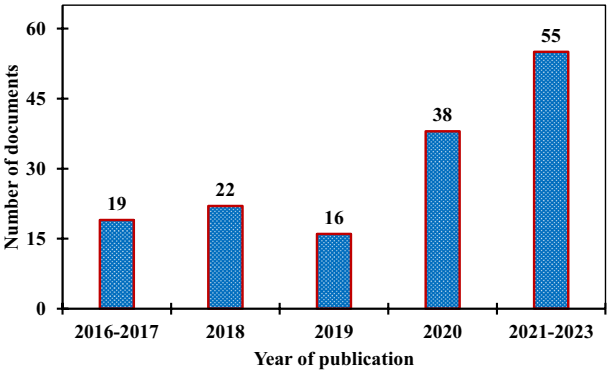


Fig. 2 Literature considered

Table 1 compares earlier reviews on PV-TE. This review details previous studies on natural, forced and hybrid cooling methods and their significant findings and challenges involved in the cooling methods of PV-TE systems. Figure 2 shows the year-wise distribution of the selected literature. Babu and Ponnambalam [13] integrated a commercial multi-crystalline PV with commercial TE on the backside of the PV made of a copper absorber plate to evaluate the theoretical performance of a hybrid PV-TE system. The power generated system was 8.3% more and the electrical efficiency was 6% greater at the standard input conditions.

This review summarizes recent research on using PV-TE with applications not represented in earlier published review articles. The paper provides a concise review

of the recent developments in PV-TE. This review briefly covers the critical analysis of the technology in its current development phase—about two-thirds of the recent literature has been published in the last three years. The innovative part of this review lies in the recent developments and indications of future solar PV-TE cooling technologies.

The present review of advanced cooling methodologies for solar PV and thermoelectric hybrid energy systems is summarized in the following sections. Section 1 introduces the sustainability and motivation of the present review on solar PV cooling technologies. Section 2 discusses the natural cooling methodologies. Section 3 discusses the forced cooling methodologies. Section 4 deals with hybrid cooling methods. Section 5 highlights the state-of-the-art solar energy harvesting techniques and discussion. Section 6 summarizes the significant conclusions and provides further scope.

2 Natural cooling

The natural cooling technique is a type of passive cooling technique. It involves reducing the temperature of the surface of the PV with the help of natural environmental factors like air, water, or other fluids without the influence of mechanical systems such as pumps, radiators, or fans. Natural cooling is one of the most economical methods since there is no extra cost for integrating a cooling technology or regular maintenance. This cooling method is amiable to the environment and does not adversely impact biodiversity. The following section discusses functioning natural cooling technology with air and water.

2.1 Air-cooling

The most abundant and prominent method used for cooling since the industrial revolution is with the help of air. Air-cooling works on heat minimization from a surface by maximizing the flow area. The amount of cooling is directly proportional to the exposed surface area. Concerning the velocity, the condition of direct proportionality stays true until a peak value is achieved, after which the rate of cooling decreases. Ahmadi et al. [21] employed paraffin as PCM to cool down PV panels. PCM was infiltrated into carbon nanotube foam to overcome the low conductivity. The energy efficiency of the leading PVT was assessed for effective, and the energy efficiency is 13–14.2% for passive and 66.8–82.6% for active cooling methods. Ajewole et al. [22] investigated the absorption of heat energy by PV solar cells to increase rooftop electricity production output PV power systems. Rooftop PV has a high potential for generating renewable electricity. Alamri et al. [23] studied the natural and forced cooling of PV-TE hybrid systems. The passive cooling of a PV-TE module is better than active cooling for a PV-TE module size and scale used in this experiment since the power demanded by the fan is higher than the power produced by the module.

Lekbir et al. [24] investigated a new PV-TE with a nanofluid cooling station to convert waste heat into electrical energy to improve the hybrid system's efficiency.

Comparative statistical studies were conducted to compare the performance of the proposed model's three configurations: concentrated photovoltaic (CPV) standalone, CPV-TE, and an integrated PVT system. Therefore, the design of the nanofluid-based CPV/TE hybrid system is proposed to improve the conversion rate of solar energy. Zhao et al. [25] studied the possibility of a PV-TE device generating electricity at night. PV-TE devices for continuous power generation day and night are proposed for deep consideration. The average output voltage of PV-TE devices was measured at 9 mV at night and 0.6 V in the daytime. Geometrical structural parameters of PV-TE leading to maximum power generation and conversion efficiency [26].

Bicer et al. [27] experimented with splitting concentrated solar light using cold mirrors for PVs and photonic hydrogen production applications. PV module's efficiency was 14% under concentration and 15.37% under no concentration. Figure 3 shows a schematic of PVs and photonic hydrogen production [28]. The incident sun energy is converged by the Fresnel lens utilized in this configuration. Spectral division with cold mirrors divides the focused solar radiation as it spreads over the photocell surface. This arrangement enables many solar spectra uses, such as hydrogen generation by photo electrochemistry. A photoelectrochemical cell is used to facilitate hydrogen production [29]. The cell consists of a membrane electrode assembly and stainless-steel electrodes. Kiyaee et al. [30] designed a stainless-steel-based linear Fresnel lens concentrator. Each lens focuses on a silicon solar cell's specific range of the solar spectrum. Due to this, decreased temperature and increased electrical efficiency were observed by eliminating high-energy solar wavelengths. With the increase in cut-in wavelength, the cell temperature is reduced, resulting in increased electric efficiency.

Meshram and Sawarkar [31] analyzed the diffuser flow for air-cooling of low-concentrated PVT solar collectors. One with curved sidewalls and three deflector plates was the most efficient among all the analyzed geometries. It is suggested that the deflector plate should be close to the inlet of the diffuser. The height of the

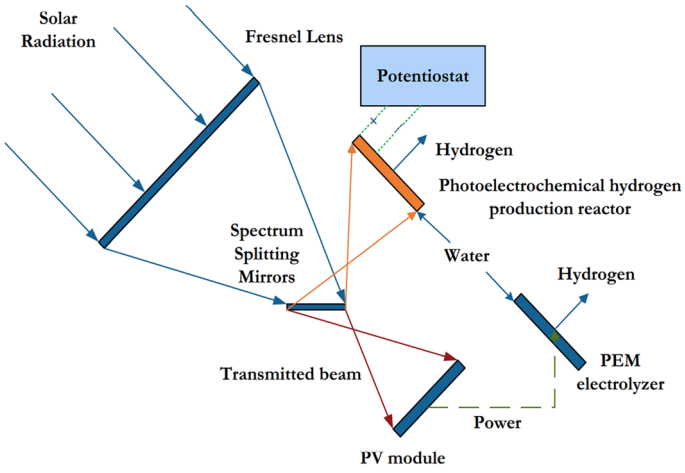


Fig. 3 Schematic of PVs and photonic hydrogen production [28]

diffuser does not affect the distribution of air velocity. The shape and position of the deflector walls do significantly affect it.

2.2 Liquid cooling

Liquid cooling encompasses using water and other liquid coolants to curtail the heat released because of the increase in the surface temperature of the PV. Liquid cooling offers a higher thermal conductivity than other available cooling methods. Nanoparticles used in nanofluids are metals, oxides, carbides, or carbon nanotubes. Nanofluids are used for their advanced thermal properties, such as cooling in heat transfer devices, heat exchangers, electronic cooling systems, and radiators. Abdelrazik et al. [32] researched the visual filtering and nano PCM on the performance of a mixed PVT solar collector. The nano PCM layer under the PV module reduces the overall performance. In addition, the refinement height and weight flow affected the thermal and electrical efficiency.

Akbar et al. [33] evaluated a hybrid PVT-TE module, including a PV module, TE module, and a cooling unit. To reduce the surface temperature, they cooled with nanofluids. The efficiency of PV depended on surface temperature. They used an optimization through which the performance of the PVT-TE system was improved. Soltani et al. [34] studied PV-TEM cooling in five different experiments starting with natural cooling. Cooling with water increased the efficiency of the system by 47.7%. Using Fe_3O_4 nanofluid to cool the module further increased the cooling than water-cooling by 5.7%. Figure 4 shows a hybrid PV-TE system with air and liquid cooling.

The ambient temperature plays a significant role in the performance of PV-TE modules. The atmospheric conditions should be high enough to complement the system but not an elevated temperature affecting the efficiency. Elevated temperatures reduce the output of PV cells due to decreased electrical energy. Al-Amri et al. [35] investigated thermal management and cooling procedures for PV panels under the harshest climate. Kidegho et al. [36] performed experiments on the performance of a PV-TE module. The cooling of the PV panels by the TE and cooling plates

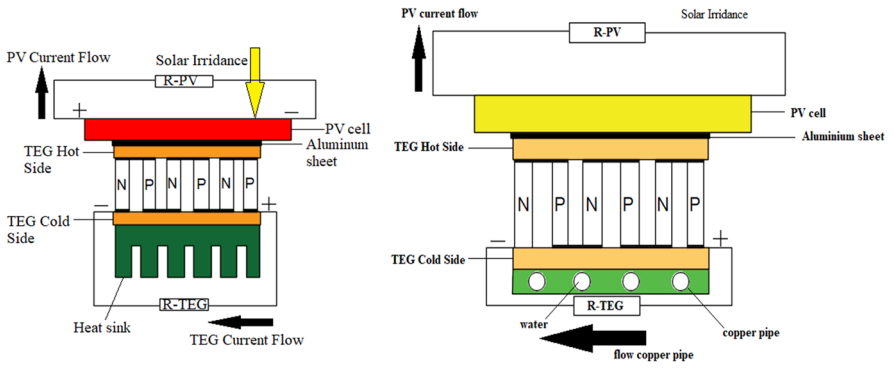


Fig. 4 Hybrid PV-TE system with **a** air-cooling, **b** liquid cooling [34]

increases the power output has been found to increase by 1%. Further, the system's performance increases by 4.42% because of the cooling effect of TE with water on the cold side.

Metwally et al. [37] developed a PV model with an efficient cooling system. The 3D model was 500 mm × 500 mm and tended to 40° angles. Various hot models are simulated using PV layers of 3D computer model, hot TE modules, and water modules. The numerical simulation effects of an efficient cooling system are compared with unpowered PV panels under forced delivery. Sabry et al. [38] investigated the PV-TE hybrid system. The performance of the PV-TE hybrid system consisted of a high-efficiency multijunction concentrator solar cell coupled to a TE module. The effect of TE size, junctions, and percentage of TE area covered was simulated. Total electrical output was measured. The performance of the system was linked to the internal structure of TE.

Yin et al. [39] compared the one-day performance of isolated PV and a TE. Low-concentration PV systems have a concentration level of fewer than ten times. Such PV systems are suitable for standalone applications and structural integration. They can capture the sun's rays effectively. Pabon et al. [40] evaluated the design parameters of the cooling. This mode benefits such as efficiency and temperature of PV cells controlled using the hidden temperature of the refrigerator, which increases electricity production without increasing the surface area of the PV cell. Zhang et al. [41] enhanced the performance of PV, TE coupling devices with thermal interface materials. The maximum power obtained was 8.9–22.76%, depending on the cooling water temperature and radiation, showing feasibility. Yu et al. [42] reviewed the research on implementing PCMs in PVT systems. Integrating PCMs can effectively upraise the performances when the phase change temperature and operation parameters are carefully altered according to environmental conditions.

Kandeal et al. [43] reviewed convective, conductive, and radiative cooling using high emissive coatings. For significant cooling following criteria should be maintained- higher radiation, a low continuous flow rate of coolant, Aero-based cooling (enhancement in electrical performance by ~ 29%), Fins attachment (enhancement in overall efficiency by 30%), and radiative cooling. Dawood et al. [44] investigated CuO nanofluid on PV-TE and determined an exergy performance of 92.47% higher than typical PV. Table 2 shows natural cooling methods.

3 Forced cooling

Forced cooling technology involves artificial methods to transfer heat from the system to the surroundings. The components include fans, water pumps, heat sinks, and TE. This method of cooling is more coherent than natural cooling.

3.1 Air-cooling

This air-cooling comprises mechanical components such as fans, condensers, and heat sinks to enhance cooling. This technique demands extra power usage in the

Table 2 Natural cooling methods

References	Nano application	Major findings
Abdelrazik et al. [32]	GO/water nanofluid	PV temperature is reduced by 16–27.6 °C. Electrical efficiency is increased by 0.87–1.49%. The thermal efficiency is increased by 11.7–17.4%
Akbar et al. [33]	SiO_2–H_2O nanofluid and Ag–H_2O nanofluid	Efficiency increased by 16.17%, and 20.68%, respectively
Soltani et al. [34]	SiO_2/water nanofluid and Fe_3O_4/water nanofluid	Efficiency improvement was 3.35% and 3.13%, respectively
Dawood et al. [44]	CuO nanofluid	Exergy generated by PV-TE is 92.47% higher than normal PV cells

form of electricity for the air-cooled channel to function effectively. Aboelmaaref et al. [45] investigated the TE air-conditioning system under hot climatic conditions in Sohag city Egypt. The system consisted of solar PV panels, a solar charge controller, a power storage system, TE, a heat sink, an inverter, a conditioned test room, a condenser, and evaporator fans. Bahtiar et al. [46] experimented with the characteristics of a PVT collector with exergy performance comparison with indoor/outdoor comparison. The accuracy of exergy and exergy efficiency was 98.42% and 98.11%.

Forced air-cooling has yet another prominent application in TE generators where the cooling is coupled with the TE to boost the cooling capacity offered to the PV system. Bamroongkhan et al. [47] assessed an experimental prototype of a parabolic PV-TE. They set a focal distance of 57 cm, which was the best focal distance for this parabolic dish. They observed that the average power of PV cells with air-cooling was 21.42% higher than that without air-cooling. Benghanem et al. [48] performed two experiments to analyze the effect of adding a TE device to a PV. The TE device was operated as a cooler to remove excess heat from the PV module. For every degree Celsius reduction in the PV module temperature, its efficiency increased by 0.5%. The additional cost of adding this system was almost 6% of the overall cost of the PV system.

Daghigh et al. [49] evaluated a dual-evaporator refrigeration system using a PV-TE solar collector. Radiation cooling has attracted a great deal of research over the years. It focused on assessing the size of the facility and the variability of cooling capacity in different areas. Combining low-cost PV with PVT can provide an exciting and inexpensive solution for radiation cooling. Lv et al. [50] proposed a novel strategy of PV-TE that enhanced sky radiative cooling. The proposed system simultaneously used PV cells to provide cold power and energy. The overall efficiency was the performance-determining factor, whose increment ranged between 34.7 and 35.7%.

Jamali et al. [51] enhanced power generation by cooling a semi-transparent PV power plant with a solar chimney. The airflow cools the PV panels on their upper sides, but the floor has no cooling flow. It was using solar chimneys for cooling improved PV power generation by 29%. Li et al. [52] examined an energy system

based on fuel cell technology and solar energy. Improving the performance of fuel cell operating temperature played a leading role. The precise adjustment was vital as it has a key role in the design, sizing and improving performance. The system's efficiency was 63.8% and the exergy was 73.01 kW. Wang et al. [53] studied the performance evaluation of a poly-generation system based on fuel cell technology and solar PV and using waste heat. The cycle comprises an alkaline fuel cell, a Stirling engine (electricity), solar PV, and an absorption chiller. The cycle produced 3.4 kW of electricity, hydrogen fuel, electrical and overall efficiencies of about 64.36 and 77.57%, and 77.44 kW of exergy destroyed.

A heat sink is a static heat exchanger that transfers the heat generated by an electric or machine to a liquid for temperature control. Megaspore et al. [54] optimized a new concept in porous media considering thermal radiation (PV panel cooling). A porous fin structure in forced convection flow for PV panel cooling was studied. The porous media increased the surface area by 18% and reduced volume by 14% more than solid fins, increasing heat transfer by 20.3%. Vittorini et al. [55] developed a mathematical model of cool PV modules and allowed room testing to improve the efficiency of non-cooling PV modules under real operating conditions. An exciting opportunity for heat recovery was established when the heat sink did not refer to the ambient air and the cooling fluid circulation. Rahman et al. [56] proposed a portable solar TE refrigerator design. The dimension of the refrigerator was 6.5 cm × 6.5 cm × 15 cm, and a storage device with a battery was incorporated. A PCM is a substance that releases/absorbs enough energy in a phase transition to provide useful heat/cooling. Rajvikram et al. [57] improved the efficiency of a solar PV panel by controlling the temperature using something called PCM. PCM HS29 was efficient in maximum power output and higher efficiency. Skovajsa et al. [58] experimented with PCM-based accumulation panels in combination with renewable energy sources and TE cooling. It uses molecular encapsulation of paraffin wax, allowing for an extremely high concentration of PCM within the polymer compound. The power balance PCM uses for cooling and heating is 1.95 and 3.71 kWh, respectively.

3.2 Liquid cooling

This liquid cooling technique incorporates water pumps, radiators, and other mechanical devices that positively affect the system's cooling. The liquid continuously passes through pathways created to extract heat from the system and through radiators or other devices where heat is removed from the liquid. Teffah and Zhang [59] proposed a PV-TE system that consists of four TE units. The system with PCM was feasible when the optimal PV operation temperature was below the melting temperature of the PCM; if the optimal temperature was higher, the newly developed PV-TE system was advised. Liu et al. [60] investigated TE and water-cooling-assisted high conversion efficiency polycrystalline Si PV. A novel PV-TE system was designed and manufactured. When the light intensities were 1000 and 800 W/m^2, the steady-state temperature of the PV-TE system was found to reduce to 32.7 °C

and 24.2 °C. Hence, the overall efficiency increased by 34.7% and 35.7%, respectively. Ren et al. [61] examined Si-based PVT systems on a stainless-steel substrate.

A Si PVT system consisted of a glass cover, a-SI cells, an aluminum absorber plate, a copper tube and an insulation layer. Fifteen pieces of a-SI cells were laminated into an absorber plate through an adhesive layer. The external environment developed and validated the distributed parameter model to analyze the performance of the a-SI PVT system at medium temperatures. Shittu et al. [62] studied a concentrated PV-TE system with and without a heat pipe. The efficiency of the PV-TE-Heat pipe was 3.31% and 58.01% more than PV-TE and PV systems, respectively, at a concentration ratio of 6, at low ambient temperature and higher wind speed. The activated alumina is produced by aluminum hydroxide by dehydroxylation to produce highly volatile materials. It has a high thermal conductivity property, which is helpful for cooling techniques. Abdo et al. [63] used saturated activated alumina with water to cool solar panels and they also aimed to assess the effect of different internal and external designs. The new internal configuration reduced panel temperature by 12–14 °C. Saline water was used as an alternative to pure water as it performed similarly. Activated alumina tablets showed a higher deterioration rate at higher salinities.

An adsorption cooling system is a heat-activated cooling system based on a solid sorption process. And it is a viable choice for solar cooling as a cooling system for absorption. Alelyani et al. [64] combined the extruded solar collector and the adsorption cooling system under a single module called the extruded solar adsorption tube module. The total uncertainty in experimental efficiency was calculated based on the Gaussian error propagation. Thus, the newly proposed evacuated tube solar adsorption cooling system was investigated. Al-Nimr et al. [65] experimented with a hybrid absorption/TE cooling system driven by a concentrated PVT unit. A large amount of cooling was obtained by the stated design conditions of the experiment when the PVT outlet temperature was maximum. To continue the efficiency process, the coefficient of performance (COP) TE-V should exceed 6.4. COP total cooling varies between 0.146 and 0.232 during the cooling period under the Mediterranean climate.

Natural air-cooling solar collectors are known for achieving better efficiency when integrated with PVT; research has been undertaken regarding forced liquid cooling. Ali et al. [66] experimented with the performance of solar ponds with PVT collectors. Solar energy is converted into a heat sink or a convective ground using a thermal coil system (fixed) and mixed PVT solar collectors. Using PV-TE has led to an increase in the efficiency of solar pool heat—15% in January, 30% in September, and 9.357% in December. Dimri et al. [67] analyzed PVT-TE collectors considering three different models—opaque, semi-transparent, and aluminum base. The analysis was based on the thermal and artificial neural network models. They found that the electrical and thermal efficiency of the PVT-TE opaque water collector was more significant than the opaque PVT-TE air purifier. They also found that water-based PVT-TE collectors' daily gains of total energy, thermal energy, and total energy efficiency are remarkably high.

Al-Nimr et al. [68] proposed a hybrid PV-TE cooler distillation system. TE is integrated to improve the performance of the PV-distillation system in tropical and

subtropical climates. The efficiency of the distillation system improved with the increase in airspeed to a certain point and then decreased slightly because of an increase in ambient temperature. The efficiency of the PV system increased with increasing wind speed. Al-Nimr et al. [69] integrated the organic Rankine cycle with PVT integrated with storage. During the summary phase, the efficiency increased with the geothermal temperature change, increasing the number of transfer units and decreasing the flow rate. Bevilacqua [70] designed a cooling system based on forced convection phenomena to enhance efficiency and limit auxiliary energy. The Internet of Things is a majorly explored topic in the present world since it bridges software-driven ideas to a mechanical background. Chawla et al. [71] studied innovative solar energy systems solutions for smart cities. A temperature-dependent Internet of Things-driven cooling unit is modeled to improve solar module output. The proposed design has a comprehensive data prediction model, simplicity, outdoor performance, and design automation.

A refrigerant is an active fluid used in the refrigeration cycle of air-conditioning systems and heat pumps where in most cases, they alternate a phase from liquid to gas and back. Daghigh and Khaledian [72] studied the characteristics of solar with a TE cooling-heating system. Results from compression cooling (R404A refrigerant) show less need for pipes and is a substitute for R134, R12, R22 refrigerants, being more economical and environment friendly. Duvenhage et al. [73] reviewed highly responsive solar and wind power plants with increased production capacity. The local water availability needs to be considered for concentrated power plant cooling. Islam et al. [74] developed an integrated system based on renewable energy to complement the energy demands of a house located off the grid. A thermodynamic software package was used for the required flexibility tests. Accuracy of thermal energy and energy efficiency was performed. Islam et al. [75] investigated a solar-based integrated system for desalination with heat recovery. Exergy efficiencies of multigeneration were 44.1–45.01%. Javidan et al. [76] investigated the thermal management of a PV using water-jet impingement cooling. The best temperature uniformity was obtained for the minimum nozzle-to-PV spacing of 5 mm, the minimum nozzle diameter of 1 mm and the maximum nozzles of twenty-four.

As discussed in the paper earlier, nanofluid has proven to increase the cooling rate of the PV system because of the improved thermal properties of cooling materials used for heat transfer. Lekbir et al. [77] compared the performance of PV-TE hybrid modules with natural cooling, water-cooling, and cooling with nanofluid. When observed with nanofluids, the module showed the least degradation in energy. Rejeb et al. [78] did comparative research on CPV-TE with 0.5% graphene/water nanofluid, water and conventional PVT collector. They used the finite volume method and performed computations. They found that the total electrical power produced on a summer day was more than that for winter. Sami et al. [79] analyzed the behavior of nanofluids in the quaternary refrigerant mixture that is PV-thermal driven and possesses cooling capability. Solar radiations heated the nanofluid and generated vapor refrigerant. The nanofluid CuO had a significantly elevated temperature leading to higher hybrid system efficiency.

The multijunction model is proposed to reduce the size of solar cells without significant losses in generating energy and overall efficiency. Li et al. [80] experimented

with thermal management's conversion efficiency gain for concentrated triple-junction solar cell systems. The output of the integrated PV-PCM-TE system was 6.16% and 1.496 W over that of the PV-TE system. The intensity of sunlight has made the output of three times more solar cells reach 4.298 W. Sajid et al. [81] made an economic comparison among four integrated systems for self-sufficient greenhouse gases driven by solar energy. The vapor compression system was more economical than the vapor absorption system. Latitude, weather, and climate are the major factors affecting the sunlight detected in a particular area. Salameh et al. [82] studied a novel three-dimensional model for the PV-T water system in hot climate regions. Simulations were performed under different irradiance conditions and high flow rates to study the performance of the PVT system. The central part of the PVT module has a remarkably elevated temperature where symmetrical boundary conditions are applied on both sides of the PVT. The irradiance and temperature difference between the ambient temperature and the penetration temperature of the cooling liquid was extraordinarily high and exceptionally low, respectively. The overall efficiency of the system improved by 1–1.5%.

Sohani et al. [83] studied the energy and exergy analyzes on a seasonal comparative evaluation of water flow cooling to improve the performance of monocrystalline PV modules in a hot-arid climate. Sharadga et al. [84] used a lithium bromide-water-based Kalina cycle for a hybrid PVT system. The overall efficiency could be improved by increasing the temperature of the generator design. The results showed the overall efficiency increased by 22–27% using the Kalina cycle. Siecker et al. [85] studied various methods to minimize the negative impacts of the increased temperature and enhance the efficiency of PV solar panels. Temperature reduction and increased efficiency were seen because of the water immersion cooling technique. The temperature problem was removed by a transparent coating that enhances the PV panel's efficiency. Song et al. [86] studied the performance comparison of direct-expansion solar-assisted heat pumps with three different PV evaporators. The refrigerant absorbs solar radiation heat and comes into the compressor by a suction inlet. The separated parallel pipelines merge. The superheated refrigerant at high pressure enters the condenser and condenses to a liquid state. Finally, it goes through the capillary and comes back to the evaporator. The thermal efficiencies of the three systems are the same, while the electrical and total efficiency of FPV–SAHP systems are higher. The total efficiencies of the three systems are 226.84%, 226.02% and 191.65%.

Maximum power point tracking refers to solar PV and wind turbines generating maximum energy regardless of weather conditions. Torbatinezhad et al. [87] analyzed cooling systems for solar panels in concentration PV systems. The PVT system was considered to have a flat linear mirror used to receive high suns. Numerical analysis was performed by modeling the geometric background and independent grid study. The effect of pin fin angles and the variability of bulk flow rate on electrical and thermal performance was studied. Wang et al. [88] designed an atmospheric plate thermosiphon cooling system that can be used for heat dissipation of single or low-concentrated solar cells. Heat resistance and start-up time, heat transfer coefficient and thermal resistance of the evaporator were investigated. Zanlorenzi et al. [89] studied efficiency improvement using a hybrid PV module through an automatic

water-cooling system. The prototype showed an average efficiency improvement of around 33.28%; applied in countries with a tropical climate, it would achieve greater efficiency and economic advantages. Han et al. [90] proposed a solar cooling system powered directly by a PV panel rather than a traditional one. The system has been evaluated for two conditions control and without. A comparison strategy of impedance under different climates was adopted and modules were designed and developed. The compressor does not stop when using the controller, while the maximum impedance is 143 W/m^2. The efficiency of solar energy conversion in the compressor, compressor utilization, compressor performance, ice quantity, and COP solar system improved by 83.7%, 79.6%, 58.1%, 81.9%, and 60.4%, respectively relation to the control system without control.

4 Hybrid cooling

Hybrid cooling amalgamates both the forced cooling techniques mentioned above to provide heat rejection and reduce water and energy utilization. This technique offers high-performance than novel cooling methods. Abdo et al. [91] studied the performance evaluation of a modern concentrator PV and solar TE generator hybrid systems design. The study was characterized by developing a novel concentrator and PV-TE modules exposed to concentrated solar radiation. The apparatus contains a microchannel heat sink attached to PV and a cold area to PV-TE. Both modules were subject to fixed irradiance with a Fresnel lens. The newly developed system maintained a temperature of 77 °C with an energy efficiency of 3.2 kW/m^2. Thus, there was an increase in the efficiency of solar cells. Lattice associated with monolithic triple-junction electrically operated and thermal PV cell concentrator with two TE Peltier modules [92]. They improved the critical thermal interfaces between the cell and TE module. The integrated TE module operates as a temperature sensor and a heat pump for the solid-state of the CPV cell. Architectures for accurate measurement of electrical devices were enabled to optimize electrical performance and system lifetime while minimizing the levelized energy cost at the system level. Arabkoohsar et al. [93] introduced a novel hybrid solar-based smart building energy system. The proposed system's technical and economic aspects were evaluated and compared with conventional setups. Fabbri et al. [94] studied the numerical modeling of the new PV-TE cooling system integrated with support. The cooling simplifies solar cell cooling and TE conversion technology by integrating a heat exchanger with a thermoelectric cooler. The apparatus (a modular structure) cools solar panels, increases efficiency, and acts as a thermal insulator for the roof. The heat sink's core constructs a compact and easier-to-build TE. For a PV panel (100 cm × 125 cm) almost 15% electrical power was seen. The Seebeck effect brings power from 61.2 to 71.2 W. Sanaye et al. [95] proposed an innovative hybrid energy system for cleaner combined cooling, heating, and freshwater production. Modeling setup was done to generate and store electricity with minimum annual cost.

Behzadi et al. [96] propose an integrated novel solar integration system to improve energy efficiency and reduce the overall cost of the system. The recently proposed model with TE instead of a condenser cooling system is very suitable. It

has a high net output/hydrogen production rate and a low charge time. The optimal performance and efficiency of the proposed system were found to be 12.01%. They also concluded that the PV/T module's location is a critical parameter and should be kept to a minimum. Contento et al. [97] conducted a study focused on enhancing the efficiency of a-Si and solar cells using various TE hybridization strategies. They did this in two ways: optically coupling the PV cell and TE to prevent physical contact and thermally coupling the PV and TE devices through direct contact. Efficiencies achieved were about 57% and 35% in thermally coupled configurations, while about 42% and 24% in optically coupled configurations. Gu et al. [98] build a mathematical model of a hybrid PV-TE system based on the thermal resistance theory of PV panel, heat sink and TE. The TE was attached to the rear side of the Tedlar by gel to form a PV-TE system. Thermal resistance analysis was done to evaluate performance.

Cui et al. [99] investigated the setting of the PV-PCM-TE compound and analyzed its performance. They concluded that this hybrid set works better than a clean PV system. This also gives the impression of improving solar energy utilization. Darkwa et al. [100] conducted a study on numerical and experimental analysis of an integrated TE-PCM power enhancement system for PV cells. The integrated PV/TE/PCM system achieved 9.5% power output, more than standard PV and PV-TE systems. Rajaee et al. [101] analyzed the effectiveness of CO_3O_4/water PCM usage and nanofluid-like paraffin wax/Alumina powder as a cooling agent for a PVT-TE hybrid setup. TE was attached to the rear side of PV. CO_3O_4/water nanofluid, two kinds of PCM, and 5K-type thermocouple sensors were also employed. Data were recorded for six days. Thus, a new setup was examined using six cooling methods. Daghigh et al. [102] compared the performance of PV panels for various cooling methods using effective liquid air and TE liquid modules under the weather conditions of Iran in September 2018. The payback period was 7.14 with a 10% correction coefficient.

Ejaz et al. [103] discussed various PV concentrators used as CPV devices. They used cheaper optical devices to enhance PV performance. With the CPV technology, electricity with high-efficiency was generated and they experienced higher temperatures which caused a reduction of PV efficiency and life span. The highest cooling effect was observed using PCM cooling technique. They found greater potential in materials for the design of more efficient PV concentrators. El-Samie et al. [104] introduced a fully integrated PVT system simulation that combines visual, thermal, and electric models on the same grid—introduced 3D digital simulation of a fully integrated PVT system that combines visual, thermal, and electric models on the same grid. They compared the hybrid PVT with a standard PV system and found that the hybrid PVT is more economically efficient. Haidar et al. [105] studied PV temperature regulation using the evaporative cooling principle and detailed theoretical and operating conditions experimental approaches. The model comprises an inclined duct with a PV panel on the upper surface. On the lower side of the duct, water flows and the air blows inside the pipe. The performance of PV panels drops with an increase in the panel temperature. A decrease in 10 °C improved power by 5%. Hammad et al. [106] proposed a cost-effective cooling method to improve PV modules' efficiency. The average efficiency enhanced using water and fins as heat sink was 5.8% and 8.17%, respectively. Kidegho et al. [107] evaluated PV cell

temperature distribution under an external setup. Under analytical evaluation, the model was used to evaluate PV cell temperature. Lashin et al. [108] implemented a CPV-TE hybrid system with a high-efficiency multijunction concentrator solar cell coupled to a TE module. The power generated by the CPV/TE hybrid system increased by 3–7.4%. Lashin et al. [109] the performance of a TE partially illuminated with highly concentrated light. A PV-TE coupled system consisted of a large Fresnel lens, cooled by attaching it to the top of the unit.

Yin et al. [110] analyzed the feasibility of tandem concentrated PV-TE. Their test device consisted of a xenon lamp, fan, reflector cup, PV-TE system, test platform, control and measurement cabinet, and computer. Liao et al. [111] ignored the system boundary heat transfer and obtained an efficiency of 20:9% at 28.7 V and 1.66 A. The analysis system's optimal performance due to solar irradiance maximum efficiency is 20:0%. Models and analysis provided various strategies for solving parametric design problems. Nižetić et al. [112] analyzed novel passive cooling strategies for freestanding Si PV panels. The highest potential of temperature reduction in PV panels was obtained with slits on the front PV panel surface and a temperature reduction of 4.0.

Shin et al. [113] experimented with hybrid PVT systems. PVT system and PV-TE harvested solar energy from solar light and cell waste heat. A thermocell consisted of electrolyte and single-walled carbon nanotube electrodes. Solar cell conversion efficiency improved from 13.2 to 15% when cooled to 34 °C and harvested an additional power of 3.53 $\mu W/cm^2$. The output power density was small, so efforts to develop efficient electrodes and electrolytes were to be made. Zhang and Xuan [114] designed an experimental setup to evaluate PV-TE hybrid system, which was installed with adjustable cooling blocks to minimize the temperature variation caused due to varying solar radiation. A cooling block controlled the cycle of cooling water with increased output power also observed. Wu et al. [115] discussed the PV and PV-TE system performance under external and internal conditions using mono and polycrystalline Si cells. The enhanced wind speed improved the performance of PV cells.

Zaghloul et al. [116] optimized and analyzed a PV cell with a high concentrator connected to the heat sink using straight wings. Solar irradiance is a direct conversion to electrical power (PV effect), dissipation by radiation and convection through the cell, and remaining transferred as heat in the solid lower parts. Heat dissipation rate depends on sink design and climate conditions. The maximum electric power of a multijunction solar cell is 2.5 W/cm^2. Increasing wind speed to 5 m/s produces electric power of 5.6 W/cm^2 and 4.2 W/cm^2 at ambient temperature. Yavuz et al. [117] studied PV and solar TE hybrid use. Using cooling water in the irrigation system, increasing the temperature between TE and surfaces was possible without phase-changing materials. Zhang et al. [118] confirmed that the performance of PV-TE hybrid devices could be improved with effective heat transfer. They removed the upper ceramic plate of TE device to make a compact PV-TE hybrid structure. Thermal greases were applied in the integrated PV-TE hybrid devices to reduce the thermal contact resistance. They also found that output power improved with a thin alumina layer. Zouak et al. [119] involved PV, Peltier and Seebeck effects in simulating a hybrid PV-TE system. To obtain the maximum power output at ambient

temperature, the device was cooled by Peltier effect and the system generated additional electric power by Seebeck effect. Table 3 shows hybrid cooling methods. Yin et al. [120] designed a thermal resistance model to predict the performance of PV-TE system. Amorphous silicon and polymer PV was preferable for the concentrating system. Adam and Ahmad et al. [121] experimented with increasing the solar system's performance by using the waste exhaust air of the centralized air-conditioning system as a cooling agent. The average temperature of the exhaust air was measured to be 28.7 °C in July. This technique increased the life span of PV panel when the operating temperature, thermal stresses and the chances of hotspot occurrence were reduced. PV panels' front and back surfaces are less exposed to additional thermal stress, so air cools them.

Al-Nimr et al. [122, 123] simulated a new hybrid PV/wind system using ducted turbine technology. To amplify the wind current, they used the buoyancy effect and the convergence duct embedded below the panels to cool the PV panels. They revealed that combining both systems showed advantages at high and low wind speeds. Al-Waeli et al. [124] improved the PV efficiency using Cobalt nanofluid and nano PCM by up to 10.05%. Asefi et al. [125] proposed plans to amalgamate PV into a building structure to raise the system's efficiency, indoor supply heating and hot water. For indoor heating, they assessed new and efficient technologies being employed to benefit from BIPV/T systems. They examined when they were operated in various conditions while improving and enhancing buildings' performance by regulating indoor temperature. Beemkumar et al. [126] found that the PV panel does not convert solar radiation into electrical energy. Some energy remains as heat, increasing the panel temperature and decreasing electrical proficiency. They also discussed the effect of an innovative cooling technique on solar PV modules by integrating PCM and TE. Elsheikh et al. [127] discovered that different solar energy systems' performance could be improved with nanofluids' help. To eradicate the problem of low specific heat of nanofluids, they used coolants to achieve higher specific values, resulting in removing more heat.

Erzen et al. [128] inspected a solar-fueled multigeneration system using TRNSYS 18 for one year. This multigeneration system could produce heating, cooling, electricity, and hydrogen from solar energy. This simulation was conducted in Izmir,

Table 3 Hybrid cooling methods

References	Nano application	Major findings
Lekbir et al. [77]	Al_2O_3 nanofluid	NCPV/TE generated 92.47% higher exergy than the standard PV cells
Rajaee et al. [101]	Co_3O_4/water nanofluid	Enhancement of overall electrical efficiency by 4.52%
Shin et al. [113]	Carbon nanotube electrodes	The conversion efficiency of solar cell improved from 13.2 to 15%
Al-Waeli et al. [124]	Nanofluid, PCM, refrigerants	Efficiency was found to be 10.05% and increase in exergy efficiency by 11.6%
Sainthiya et al. [133]	Water nanofluid	Electrical efficiency 18.83%, Thermal efficiency 43.8%, overall efficiency 64.56%

Turkey. Gharzi et al. [129] observed increased electrical efficiencies and reduced PV cell damage when they cooled PV and CPV modules. They also observed that jet intrusion and combined methods are the finest cooling methods in any climatic condition. Maghrabie et al. [130] studied the characteristic effects of PVs because of the increase in surface temperature of the PV panels. The average overall efficiency achieved was 48.7% in April and the maximum average thermal efficiency was 37%. Manasrah et al. [131] analyzed the forced cooling of a PV module with 5 DC power fans and different cooling paths. Cooling paths offered a significant decrease in the temperature of up to 27.5% and 32.8% and a uniform temperature distribution. While cooling paths offered 12.5% and 17% with a nonuniform temperature distribution. The similarities between the indoor/outdoor conditions with various parameters such as electrical and thermal efficiency and power output are discussed. Ghadikolaei et al. [132] explored enviro-economic review of PV cells cooling technology's effect on CO_2 emission reduction. It shows that generating electrical energy from fossil fuel generators emits more CO_2 than PV panels. Nanofluids based cooling is effective in cooling PV-TE systems [133]. Salvi et al. [134] reviewed the issues related to solar receivers. Solar selective coating, absorber geometry, glazing and heat transfer fluid developments, the role of heat transfer fluid and solar tracking systems are the key issues in solar receivers. PV-TE systems are beneficial for producing hot/cold water and the maximum COP of 1.73 was attained at the maximum water flow rates [135]. The water temperature reached 49 °C. Tan et al. [136] conducted a comparative case study of remote area power supply systems using PV battery vs TE-battery configuration. The sizing and cost breakdowns of TE and PV battery systems revealed that the former is 1.66 times more expensive. Hybrid cooling is one of the effective cooling methods [114, 137]. Thus, the optimal blending of different energy systems might be effective for future sustainable power development.

5 Discussions on state-of-the-art solar energy harvesting technologies

Hossen et al. [138] compared the different dispatch methods for an off-grid hybrid microgrid system in a remote location. This consists of a combination of PV, diesel generators, wind turbines and battery storage. From a comprehensive experiment, the load-following strategy has the best performance regarding carbon dioxide emission, net present cost and energy cost. Out of the other investigated methods, namely charging cycles, generator order and combined discharge, the latter produced the least efficiency with the highest carbon dioxide emission, net present cost and energy cost. The system proved ideal for locations with similar meteorological conditions, promising good reliability. Kumar et al. [139] similarly adopted two dispatch strategies, load-following and cycle charging, to optimize the study on three different battery chemistries. The higher efficiency was recorded with the reference system following the load-following method utilizing the nickel–iron battery. A sensitivity analysis was performed using biomass collection rate, interest rate, and diesel prices. Muppidi et al. [140] focused on improving the line load ability of

transmission systems by considering the standard voltage stability index. The problem of optimal load ability was considered using the gray wolf algorithm. Shezan et al. [141] evaluated five different dispatch strategies, considering techno-economic and system stability analysis parameters. The different strategies evaluated were load-following, Combined dispatch, a predictive strategy, generator order, and cycle charging dispatch strategy. The load-following strategy was the best approach for the hybrid renewable energy systems design. The combined dispatch technique was observed to discharge considerable amounts of carbon dioxide, net present cost and system stability performance.

Shezan et al. [142] evaluated the optimum design of a freestanding hybrid microgrid for multiple unique dispatch controllers. The reactions of the power system were investigated, and a reliability analysis was carried out. The load-following strategy was found to be best in terms of the stability and reliability of the system. This hybrid micro-off-grid energy optimization model could be ideal for isolated areas. Shezan et al. [143] optimized an off-grid microgrid hybrid system sizing for four different dispatch strategies. The voltage, frequency response and sizing of the load-following type of dispatch strategy are optimum. The economic and environmental impacts of the load-following strategy were the most efficient. The system was proposed for the island and isolated areas. Farh et al. [144] applied a novel bonobo optimizer technique to find the optimal design for an off-grid system. According to the simulation results, the bonobo optimizer outperformed the other four metaheuristic algorithms.In comparison to the other four metaheuristic optimization techniques, these findings demonstrate the effectiveness and robustness of the proposed bonobo optimizer approach. Samy et al. [145] established an innovative statistical and neural network-based decision-making tool to evaluate hybrid system optimization tactics. Cai et al. [146] reviewed a variety of complications with hybrid systems. System configurations, energy storage alternatives, modeling, and artificial intelligence-based optimization were covered extensively. Innovative solutions are required to improve the sustainability of using solar energy systems towards a circular economy [147–149].

The essential three performance-determining factors are the overall efficiency, exergy, and COP of the respective systems. PV-TE produces hot/cold fluids in addition to electricity generation. Natural cooling is observed to be cost-competitive. Passive cooling can enhance energy efficiency by up to 15%. Natural cooling is preferable for small-scale solar PV-TE systems due to less input energy. Sky radiative cooling can produce the overall efficiency of PV-TE systems by about 35.7%. Using lower melting temperatures, PCM effectively cools the PV panels. Such PCM systems work passively without additional power input. Forced cooling is found to be more efficient but not cost-competitive. The energy efficiency of active cooling of PV-TE can attain about 82%. Hybrid cooling methods are effective because of polygonation using PV-TE. Hybrid cooling can enhance the energy efficiency of PV-TE by up to 55%. Using selective PCM helps improve the overall efficiency of the PV-TE system. PCM and nanofluids improve PV-TE systems' productivity through passive and active methods. Nanocomposite PCM is one of the effective passive coolers of PV-TE. The active and passive methods of cooling PV-TE are to be designed as per the applications. Figure 5 shows the efficiency of hybrid cooling.

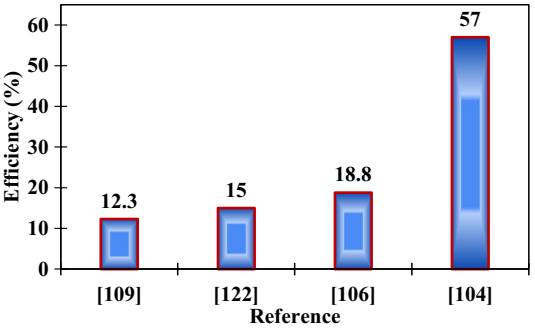


Fig. 5 Comparison of achieved efficiency using hybrid cooling methods

Table 4 shows a comparison of cooling methods with performance and maintenance measures. Teffah et al. [62] developed a hybrid system consisting of three thermoelectric components: a thermoelectric generator (TEG) and a triple-junction solar cell. Gu et al. [104] developed a hybrid system consisting of a PV panel and TEG. A TEG was attached to the tedlar's reverse side to create a PV-TE system. Darkwa et al. [106] developed a hybrid system with TEG modules with a PCM layer attached to the reverse side of the PV panel.

Ejaz et al. [109] discussed and worked on the concentrated PVs technique, which uses concentrating solar light on some areas of the solar cell. PV cells directly convert solar energy into electricity. Salameh et al. [120] performed computational analysis for the analysis of PVT hybrid systems. The most practical solution from a technical and financial standpoint would be passive air-cooling with fins. PCM is commercially feasible for passive cooling or other active cooling strategies. A water-based active cooling system could provide an economically feasible option depending on how well the system is optimized and how the process is controlled.

Future technological advancements should concentrate on creating hybrid cooling techniques to keep the surface temperature low and steady. One of the two potential cooling methods, active water or combined heat pipe and sink cooling, should be the focus of future studies. It should concentrate on the economic and environmental effects of cooling systems to close the research gap and create useful, sustainable PV systems. The need for original research data on PV deterioration, long-term advantages, the effects on operation and maintenance, and the overall durability of passive cooling systems must also be emphasized. The review in this work could serve as a helpful basis for future study.

There are significant impacts of heat on PV electrical performance. Much work must be done to find efficient techniques to cool PV modules. Recognizing and comprehending the many factors influencing cell temperature is necessary to understand how prospective cooling solutions can be impacted. Designing a system that considers various variables, including module orientation, placement, and cooling system components, is crucial. It is necessary to balance improved performance and a higher initial cost. Because it does not increase the cost of the system, any successful cooling system must be very affordable. Managing the large surface area that must be cooled while considering the low power output per module is necessary.

Table 4 Comparison and summary of cooling methods

Title	Cost	Maintenance measures	Efficiency
Natural air-cooling	No major costs involved	No special maintenance is required	This strategy has the lowest efficiency compared to the other two cooling techniques
Forced air-cooling	The cost includes the mechanism cost, such as a fan and the electricity cost	The system demands minimal maintenance with occasional dust cleaning	Air cooling is less effective than liquid cooling due to low thermal conductivity
Forced liquid cooling	The cost includes the initial equipment needed, such as a hydraulic pump, fluid lines, and a heat sink—a high cost and maintenance	The frequency of maintenance and costs are comparatively higher due to multiple components exposed to harsh weather	Due to the higher thermal conductivity of liquids, liquid cooling is proven more effective than air cooling
Forced hybrid cooling	The cost includes the air and liquid systems—high initial investment	The hybrid cooling system has the highest maintenance, consisting of several components extracting heat	Hybrid cooling is more effective than natural cooling but less effective than forced cooling

The expenses of device maintenance may outweigh the advantages of increased power production if a system is constructed without considering environmental factors. These issues must be considered when constructing a PV cooling system if the cooling is to be maximized. In conclusion, several difficult issues are associated with efficiently cooling PV modules, which have prompted ongoing study and potential for further advancement.

6 Conclusions

The present review focused on various cooling technologies used for PV-TE hybrid systems. The salient features of each cooling technology are critically analyzed for the betterment of the solar community.

6.1 Main conclusions

Based on the critical review, the major conclusions are drawn and stated below.

The key benefit of passive cooling techniques is their independence and lack of additional power consumption while retaining a straightforward design.
Active methods are preferable when a better cooling rate and high-performance heat transfer are required.
The design of the cooling channels, their mass flow rates, and the kind of coolant employed significantly impact the active cooling techniques' effectiveness.
By mounting fins on the module's back, passive cooling may efficiently lower operating temperatures while increasing power output and efficiency. However, the heat transfer area and wind speed have the most effects on how well they operate.
Due to reduced operational costs, passive cooling is more cost-effective than active cooling, although active cooling also results in a more significant temperature decrease in the PV cells.
Compared to pure base fluid, using nanofluid in hybrid PVT systems substantially impacts PV cell cooling and raises the panel's thermal efficiency.
In the PCM cooling technique, the environment's weather should be considered when choosing the PCM type since improper PCM results in ineffective cell performance and cooling. The lifespan of a PV panel is shortened.
Nanofluids can improve electrical, thermal, and overall system efficiencies while being used as spectral splitting in PVT systems.
Due to the increased thermal conductivities of nanofluids, the nanofluid PVT system has more substantial heat and electrical efficacy, resulting in greater heat transfer characteristics than the primary fluid.
Combining active and passive cooling systems with PV panels to form a hybrid solar PVT system has enabled substantial progress in enhancing electrical efficiency.

Forced cooling systems are more efficient by almost 30% than natural ones but are not cost-effective.
By applying forced cooling, the combined system of PV-TE-PCM increased the output by almost 50%.
Nanofluid in the cooling complemented the exergy generation by almost 90% compared to standard PV modules.

6.2 Future scope

The presented review on PV-TE systems shows that a combined cooling system with selective heat recovery can enhance the system's overall energy and exergetic performance. The further scope of the PV-TE cooling systems is as follows.

- The liquid cooling system with nanoparticles requires a through-study on trade-off between the cost-effectiveness of the pumping power and pressure drop.
- Using PV-TE, hybrid power generation requires more productive materials research and economical mass production.
- The compatibility of PCM and nanomaterials are to be investigated thoroughly to improve the overall productivity of PV-TE hybrid systems.
- The sizing of the additional circulation setup is crucial to improve the overall system's economics. Hence, the highly conductive thermal masses are essential to minimize the power consumption by the blower/pump.
- The proper selection of heat transfer fluids is essential to improve thermo-hydraulic performance.
- The higher temperatures of CPV technology require robust cooling technology, especially forced liquid cooling. Hence, the tribological properties of the fluids are to be investigated to improve the life of the system components.

Thus, the further scope of PV-TE cooling methods is futile and highly demanding. Ongoing research should be strengthened to meet sustainable development goals through advanced materials research in this context.

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Declarations

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